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September 11, 2026

1. Major Engineering Updates

2. Han Full Campaign: Interim Status

3. Theoretical discussion: Log-Barrier

Major Engineering Updates I

Data, Cache, and Memory Management.

- Construct the data needed for current train-val, save as a local file. (Peak RSS 75% less, loading time 11 $\rightarrow$ 1.85)
- Cache certain data, distribute to all algorithms as needed, avoid repeated computation or storage.
- Use memory-mapped files to store some precomputed static features, eliminate the copy of large data.
- Improve memory management strategy, reduce full-scale memory scan.
- Calculating optimization day-to-day, rather than waiting for the whole days' results.
- Timely cache release.

Major Engineering Updates II

Prediction and Optimization Improvements.

- Change from CPU $\rightarrow$ CVXPY $\rightarrow$ Clarabel to directly CPU $\rightarrow$ Clarabel. (15.66s $\rightarrow$ 7.78 s / 600runs)
- Opt. problem reuse. E.g., SPO+, each run solves the same problem with only different $\hat{y}$ or y_{true} . Thus, reuse it for Clarabel: `solver = build_solver(P, A, b, cones, q_pred); result_pred = solver.solve(); solver.update(q=q_true); result_true = solver.solve().` (6% less time)
- Avoid repeated computation of the same $\hat{y}$ or y_{true} in the same batch for IC, MSE and decision regrets.

Major Engineering Updates III

Validation and Testing.

- Use sampling validation. From 240 days to only 48 days for validation (for early-stop criteria).
- Reduce auxiliary feasibility check. E.g., for Tracking Error $TE(w) = \sqrt{(w - w_{\text{bench}})^T \Sigma (w - w_{\text{bench}})} \leq 4\%$, we do conduct a pre-check for the feasibility for 4% TE. Pre-checking is somehow reasonable to make sure, e.g. SPO $w^*(\hat{y})$, $w^*(y_{\text{true}})$ solving the same problem rather than relaxing differently. Yet we try to make this procedure efficient.

Window	# Stocks	Before	After	Speedup
2011-06-30	903	0.487 s	0.028 s	17×
2017-12-29	2,741	3.908 s	0.091 s	43×
2023-12-29	4,786	13.388 s	0.173 s	77×
2025-12-31	4,866	14.397 s	0.176 s	82×

Major Engineering Updates IV

Several Takeaways.

- CUDA is needed.

Method	Full CPU	C+G	Reduction
PTO-MSE	56.585 s	37.559 s	33.62%
SPO+	103.634 s	93.153 s	10.11%

- CuClarabel may not outperform. Precision is fine, but speed is not. (e.g., ablation study: 6.64x slower v.s. CPU on HS300, 2.27x slower on CSI1000) (But yet haven't been tested on real pipeline, may be different)

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Han Full Campaign: Sealed Training Coverage

- Protocol: nine annual refits (2017–2025), three random seeds, and two training objectives: PTO–Pearson IC and SPO+. The planned training total is therefore $9 \times 3 \times 2 = 54$ units.
- Campaign snapshot: 52 units have sealed `summary.json` artifacts. These are the only units summarized on the following slide.

Refit windows	Refit dates	Sealed units	Status
w_0-w_7	2017–2024	$8 \times 3 \times 2 = 48$	complete
w_8	2025-12-31	4/6	seed 3 remains in training
Total	2017–2025	52/54	interim snapshot

Scope of this update

At the training snapshot above, no OOS decisions, return series, transaction costs, or cross-window performance aggregates were available. See the subsequent OOS launch register.

Sealed Units: Training and Validation Diagnostics

Values are descriptive summaries of sealed training artifacts; validation regret is used for checkpoint selection and is not an OOS performance result.

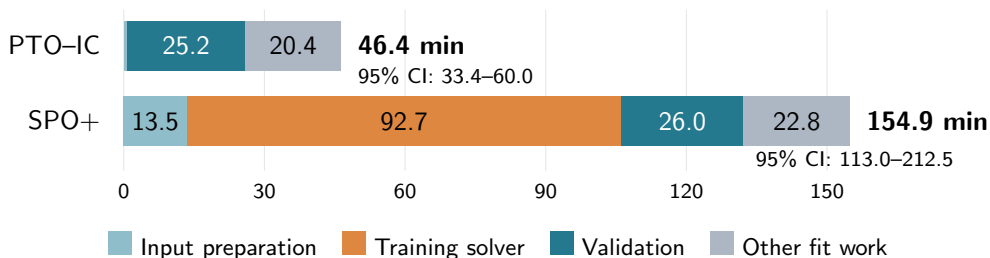
Objective	Units	Median selected epoch	Median stop epoch	Median validation regret	Cumulative fit time (h)
PTO–Pearson IC	26	11.5	26.5	1.2073	20.1
SPO+	26	12.5	27.5	1.1306	67.1
All sealed units	52	12.0	27.0	1.1457	87.2

- Each sealed unit has 48 effective validation dates, zero recorded validation failures, and a valid selection record (2,496 validation dates in total).
- The artifacts record 496,028 SPO+ training-oracle solves. Across all sealed units, 2,622 solver recoveries were recorded and zero unrecovered solver failures were recorded.
- The reported fit time is the sum across units, rather than elapsed wall-clock time for the campaign.

Where Does the Fit Time Go?

Average time per completed fit, in minutes

26 fits per method



SPO+: 60% in training solves

The largest measured component is the portfolio solver.

PTO-IC: 54% in validation

Validation takes about 25 minutes for either method.

95% percentile CIs: 20,000 refit-window cluster bootstrap draws (9 windows, 26 fits/method). Fits have different stopping epochs. These intervals describe between-window variation; overlapping windows limit independence.

Fit Time by Stage: Mean and 95% Confidence Interval

Minutes per completed fit

PTO-IC: 26 fits SPO+: 26 fits

Stage	PTO-IC	SPO+
Data / feature construction	0.74 [0.55, 0.93]	0.74 [0.52, 0.98]
Optimization problem construction	–	12.72 [9.19, 17.77]
Prediction (forward pass)	4.43 [3.24, 5.84]	4.71 [3.39, 6.95]
Backpropagation	11.27 [8.24, 14.82]	12.60 [9.09, 18.32]
Training solver	–	92.71 [69.36, 124.38]
Parameter update / gradient handling	0.15 [0.13, 0.18]	0.15 [0.13, 0.19]
Validation (total)	25.17 [17.35, 33.17]	25.98 [16.74, 38.07]
Other fit work	4.58 [3.56, 5.53]	5.34 [4.20, 6.48]
Complete fit	46.35 [33.42, 60.00]	154.94 [113.03, 212.46]

Training phases include nested work; validation uses wall time. Other = remaining fit time. “–” = not applicable. 95% CIs: 20,000 window-cluster bootstrap draws, seeds retained. Descriptive; windows overlap. CI endpoints are not additive.

Superseded OOS: Execution Bug Identified

Same seeds for both methods: 1 and 2. Each seed covers the full OOS calendar. Statistics below summarize the two seed-level metrics.

Method	Annualized net excess return			Median IR	Median TE
	Median	Mean	Range		
SPO+	3.70%	3.70%	2.53–4.88%	0.58	6.20%
PTO–IC	2.71%	2.71%	1.87–3.54%	0.39	6.85%

Withdrawn as a strategy comparison; corrected OOS running

Median net excess return: **3.70% for SPO+ vs. 2.71% for PTO–IC.**

Median paired difference: **+1.00 percentage point per year.**

Seed-level direction is mixed: SPO+ leads in one of the two paired seeds.

With two seeds, the median equals the mean. PTO seed 3 is not finished, so SPO+ seed 3 is excluded from both aggregates. Frequent no-trade decisions and carried valuations remain part of these interim results.

Superseded OOS: Paired Diagnostics

Annualized net excess return versus 000985. Each completed chain covers 404 decisions and 2,020 daily returns.

Seed	SPO+	PTO-IC	Difference (pp)	SPO+ IR	PTO IR
1	4.88%	1.87%	+3.01	0.72	0.35
2	2.53%	3.54%	-1.02	0.45	0.42
Mean (1-2)	3.70%	2.71%	+1.00	0.58	0.39

Diagnostic values only; corrected OOS running

SPO+ is ahead by 3.01 percentage points in seed 1 and behind by 1.02 points in seed 2. The paired mean difference is +1.00 point per year.

Seed 3: SPO+ complete; PTO training unfinished, so it is excluded from both method means. Means are averages of seed-level metrics, not an ensemble. Frequent no-trade decisions and retained asset valuations limit interpretation.

Superseded OOS: Risk and Execution

Both methods use the same calendar, benchmark, fee policy and missing-exit-quote carry rule.

Seed	Method	Realized TE	Excess max DD	Fee drag (pp/yr)	No-trade
1	SPO+	6.79%	-9.03%	0.09	383/404
1	PTO-IC	5.35%	-9.19%	0.59	340/404
2	SPO+	5.61%	-7.09%	0.09	383/404
2	PTO-IC	8.35%	-16.40%	0.17	383/404

- Large numbers of no-trade decisions retain earlier portfolios. These are conditional backtest results, not evidence of normal rebalancing.
- Missing stock close returns retain the previous asset valuation. Max DD is measured on relative wealth versus the benchmark.

Training and OOS Runtime: Matched-Seed Comparison

Training: sum over annual refits. **OOS:** one continuous evaluation chain.

Seed	Training (hours)		OOS (minutes)	
	SPO+	PTO-IC	SPO+	PTO-IC
1	22.37	8.20	3.16	7.37
2	19.59	6.19	4.83	8.66
3	29.04	5.69 (8/9)	5.81	–
Median (seeds 1–2)	20.98	7.20	3.99	8.02

Training takes hours; full OOS evaluation takes minutes. Training includes preparation and validation; PTO seed 3 has eight of nine refits sealed. All five OOS chains are complete (CPU inference, 404 decisions).

Matched-seed means equal medians here. Artifact preflight is excluded. Different no-trade counts affect OOS runtime; these are not per-solve timings.

Log-Barrier: Matched-Prefix OOS Diagnostic

Seed 1, w_0-w_2 : 146 decisions and 730 common daily returns (2018-01-05–2021-01-06).

Method	Net cumulative	Annualized relative	TE	IR	Net MDD
Log-barrier	16.98%	0.44%	2.30%	0.14	-31.58%
SPO+	29.54%	4.04%	5.56%	0.74	-29.93%
PTO–Pearson IC	20.97%	1.61%	5.73%	0.31	-33.83%

Method	Training total (h, w_0-w_2)	OOS total (s)	OOS mean (s/decision)
Log-barrier	12.24	262.35	1.797
SPO+	4.43	20.07	0.137
PTO–Pearson IC	1.63	53.88	0.369

The common benchmark cumulative return is 15.51%. The barrier chain records 125 carried-restricted-exit, 17 optimal, and 4 optimal-inaccurate decisions; no solve failure or warning is recorded. OOS timing uses CPU inference on one local host; the barrier mean is $13.07\times$ SPO+ and $4.87\times$ PTO–Pearson IC.

Training timing is descriptive: the barrier fits ran on a GTX 1660 Ti server, whereas the peer fits ran on a local RTX 4060. This is a matched-calendar prefix diagnostic, not a final full-sample strategy comparison: the artifacts share source, dataset, and problem identities, but have different campaign protocols and calculated data identities.

OOS Evaluation: Launch Register

Interim evaluation requested on 11 September 2026.

Five complete annual checkpoint chains are selected from the existing fits.

Method	Seeds	Decisions per chain
SPO+	1, 2, 3	404
PTO–Pearson IC	1, 2	404

- Decision dates: 2018-01-04 to 2026-04-29, every five trading days. Holdings remain continuous across annual refits.
- CPU inference and Direct Clarabel; existing artifact identities, portfolio constraints and transaction-fee policy are retained.
- PTO seed 3 remains in training. The initial five-chain subset does not constitute acceptance of the full six-chain campaign.

Output: `results/han_full_v7_interim_oos_20260911`.

Status: all three SPO+ chains completed 404/404 decisions.

PTO seeds 1–2 completed; paired interim comparison is available.

OOS Missing Quotes: Continue Without Trading

Observed issue. At the first missing-quote stop, 40 excluded holdings lacked execution quotes. The source contains no market rows for these stocks around that date; 38 have no records after 2021-08-25.

- Existing positions can remain after earlier trading restrictions; leaving the optimization universe does not itself liquidate a position.
- Adopted interim rule: retain all holdings and skip that rebalance when an exit quote is missing. Record the date, affected securities, and weights; retry at the next scheduled decision.
- No turnover fee is charged for the skipped trade. Missing stock close returns retain the previous asset valuation; other holdings continue to be valued normally.

How the results will be reported

Disclose missing-quote carry counts and affected holdings alongside returns. Retained valuations are assumptions, not observed transaction prices. The market-data files alone do not establish the corporate-event cause.

Remaining Gates Before Results Reporting

1. Seal the remaining w_8 , PTO seed-3 training unit.
2. Produce OOS decisions under the fixed portfolio and numerical policy.
3. Aggregate OOS return, turnover, transaction-cost, risk, and constraint diagnostics across all annual refits and seeds.
4. Check campaign identity and completeness before making any comparative performance statement.

Interpretation

The present evidence supports a progress and numerical-stability update only. It does not support a conclusion about economic performance or the relative effectiveness of PTO–Pearson IC and SPO+.

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Pipeline Recap: Prediction Module

- At day t , given 30 historical days' data, totally m_t stocks, each has six features, we have feature tensor $\mathbf{X}_t \in \mathbb{R}^{m_t \times 30 \times 6}$ ¹, and the corresponding label $\mathbf{y}_t \in \mathbb{R}^{m_t}$.
- For stock i at day s , let $r_{i,s} = \text{adjclose}_{i,s} / \text{adjpreclose}_{i,s} - 1$. Then accumulate for the next future 21 days: $R_{i,t} = \prod_{k=2}^{22} (1 + r_{i,t+k}) - 1$. Then 3IQR-standardize + normalize² twice, and save as float32. And this is the label $\mathbf{y}_t$.
- Then use a GRU to predict each stock a score according to the past 30 days' features, denoted by $\mathbf{s}_t := f_\theta(\mathbf{X}_t) \in \mathbb{R}^{m_t}$, where $|\theta| = 62976$. And totally, there are approximately 2380 trade days each window, i.e., $t \in \mathcal{T}_{\text{train}}$ and $|\mathcal{T}_{\text{train}}| = 2380$. For each epoch, we randomly sample 200 days (i.e., 200 different t 's).
- Then normalize $\hat{r}_{i,t} = \frac{s_{i,t} - \bar{s}_t}{\text{std}(\mathbf{s}_t)}$, and this is the input of the optimization problem.

¹Technically, all prices are divided by the closed price at trade day t , and similar to turnover

²Given $R_{i,t}$, 3IQR: denote $\text{IQR} = Q_{.75} - Q_{.25}$, then clip $R_{i,t}$ to $[Q_{.25} - 3\text{IQR}, Q_{.75} + 3\text{IQR}]$, then normalize by $(R' - \text{mean}(R')) / \text{std}(R')$.

Pipeline Recap: Optimization Module

Suppress the time subscript. Let N denote the number of stocks in the current optimization universe, $\hat{r} \in \mathbb{R}^N$ the standardized predicted scores, and $w_{\text{prev}}, w_b \in \mathbb{R}^N$ the previous portfolio and benchmark weights. Define

$$F \in \mathbb{R}^{N \times 39}, \quad \Sigma_f \in \mathbb{R}^{39 \times 39}, \quad D = \text{diag}(v) \in \mathbb{R}^{N \times N}, \quad Q = F \Sigma_f F^\top + D,$$

where F contains stock risk-factor exposures, Σ_f is the factor covariance matrix, and D contains idiosyncratic variances. The vector $\mathbf{1}_B$ indicates benchmark constituents.

$$\begin{aligned} \max_{w \in \mathbb{R}^N} \quad & \hat{r}^\top w - c \|w - w_{\text{prev}}\|_1 \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \\ & w^{\min} \leq w \leq w^{\max}, \\ & f^{\min} \leq F^\top (w - w_b) \leq f^{\max}, \\ & (w - w_b)^\top Q (w - w_b) \leq \sigma^2, \\ & \mathbf{1}_B^\top w \geq \beta. \end{aligned}$$

Optimization Module: Conventions

- $N = m_t$ for training and labeled validation cross-sections. The OOS optimization universe is not filtered by future-label availability. Throughout these slides, $F \in \mathbb{R}^{N \times 39}$; the implementation stores its transpose.
- $w^{\min}, w^{\max}$ are effective bounds incorporating nonnegativity, the 9% stock-weight cap, benchmark-relative bands $w_b \pm 0.02$, and price-limit/suspension adjustments. Since $w^{\min} \geq 0$, a separate $w \geq 0$ constraint is redundant. Do not add another barrier for this redundant constraint: it changes the objective at finite barrier parameters.
- Treat σ as the given final risk limit for the current optimization, after the prescribed TE adjustments. The base annual TE limit is 4%, with $\sigma^2 = \sigma_{\text{TE,year}}^2/12$ for the applicable annual limit.
- Full turnover is $\|w - w_{\text{prev}}\|_1 + T_{\text{out}}$. The objective omits the constant $-cT_{\text{out}}$; retain it when computing full utility values.
- Defaults: $c = 0.8$, $\beta = 0.8$; active style/industry exposure bands: 0.2/0.02.

Barrier Problem Construction

Some preparations:

- Introduce the following slack variables: $s_w^- := w - w^{\min}$, $s_w^+ := w^{\max} - w$, $s_f^- := F^\top(w - w_b) - f^{\min}$, $s_f^+ := f^{\max} - F^\top(w - w_b)$, and $s_{TE} = \sigma^2 - (w - w_b)^\top Q(w - w_b)$, $s_B := \mathbf{1}_B^\top w - \beta$.
- First the ℓ_1 term is smoothed by: $\Phi_\delta(w) := \sum_{i=1}^N \left[\sqrt{(w_i - w_{\text{prev},i})^2 + \delta^2} - \delta \right]$.
- The log-barrier term is: $\mathcal{B}(w) := \sum_{i=1}^N \left[\log(s_{w,i}^-) + \log(s_{w,i}^+) \right] + \sum_{j=1}^{39} \left[\log(s_{f,j}^-) + \log(s_{f,j}^+) \right] + \log(s_{TE}) + \log(s_B)$.

Then the barrier problem is:

$$\begin{aligned} \max_{w \in \mathbb{R}^N} \quad & \hat{r}^\top w - c \Phi_\delta(w) + \mu \mathcal{B}(w) \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \quad s_j(w) > 0 \text{ for all } j. \end{aligned} \tag{1}$$

Here $s_j(w)$ denotes the slack variables defined above, and $\mu > 0$ is the barrier parameter.

Backward Pass: What Are We Differentiating?

The forward pass produces a portfolio. The backward pass asks: how should the predicted scores change to improve that portfolio?

$$\theta \longrightarrow \hat{r} \in \mathbb{R}^N \longrightarrow w^*(\hat{r}) \in \mathbb{R}^N \longrightarrow \mathcal{L} \in \mathbb{R}.$$

Let $y \in \mathbb{R}^N$ be the observed label vector. Define the original portfolio utility and its best feasible portfolio by

$$U_y(w) = y^\top w - c\|w - w_{\text{prev}}\|_1, \quad w_y^* \in \arg \max_{w \in \mathcal{W}} U_y(w),$$

where $\mathcal{W}$ is the original feasible set. The training loss is

$$\mathcal{L} = U_y(w_y^*) - U_y(w^*(\hat{r})), \quad g_w := \frac{\partial \mathcal{L}}{\partial w^*} = -y + c \operatorname{sign}(w^* - w_{\text{prev}}).$$

The upper loss keeps the original L1 turnover; smoothing is used in the lower problem. At zero turnover, use the subgradient 0. The common constant $-cT_{\text{out}}$ cancels in this regret.

For this backward pass, hold y , w_{prev} , the constraints, μ , and δ fixed. The oracle term is then constant with respect to $\hat{r}$; gradients do not continue into previous trading dates.

Backward Pass: Start from the Optimality Conditions

Write the same barrier problem as a minimization:

$$\psi(w; \hat{r}) = -\hat{r}^\top w + c\Phi_\delta(w) - \mu\mathcal{B}(w), \quad \min_{\mathbf{1}^\top w=1} \psi(w; \hat{r}).$$

All barrier slacks must be positive. Let $\nu \in \mathbb{R}$ be the multiplier for the sum-to-one constraint. At the solution,

$$\nabla_w \psi(w^*; \hat{r}) + \nu^* \mathbf{1} = 0, \quad \mathbf{1}^\top w^* = 1.$$

If we slightly change $\hat{r}$, both w^* and ν^* change. Differentiating these two equations gives

$$\underbrace{\begin{bmatrix} H & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{bmatrix}}_{K \in \mathbb{R}^{(N+1) \times (N+1)}} \begin{bmatrix} dw^* \\ d\nu^* \end{bmatrix} = \begin{bmatrix} d\hat{r} \\ 0 \end{bmatrix}, \quad H := \nabla_w^2 \psi(w^*; \hat{r}).$$

The first row describes how the optimum moves. The second row ensures that the change in weights still sums to zero.

Backward Pass: Where Does the Curvature Come From?

The predicted-return term is linear in w , so its Hessian is zero. Curvature comes from smoothed turnover and the barrier. With $d_i = w_i^* - w_{\text{prev},i}$,

$$c \nabla^2 \Phi_\delta(w^*) = c \operatorname{diag} \left(\frac{\delta^2}{(d_i^2 + \delta^2)^{3/2}} \right).$$

For each linear slack s_j , let $a_j = \nabla s_j$. Its contribution to H is $\mu a_j a_j^\top / s_j^2$. Let $\mathcal{A}$ collect these linear slacks and set $x = w^* - w_b$. Including the quadratic TE slack gives

$$H = c \operatorname{diag} \left(\frac{\delta^2}{(d_i^2 + \delta^2)^{3/2}} \right) + \mu \sum_{j \in \mathcal{A}} \frac{a_j a_j^\top}{s_j^2} + \mu \left(\frac{2Q}{s_{TE}} + \frac{4(Qx)(Qx)^\top}{s_{TE}^2} \right).$$

All terms are evaluated at w^* . As a slack gets small, its curvature grows: moving towards that boundary becomes harder. For $c > 0$, $\delta > 0$, and $Q \succeq 0$, H is positive definite at a strictly feasible point.

Backward Pass: One Linear Solve Gives the Gradient

We already have $g_w = \partial \mathcal{L} / \partial w^*$. We need its effect on the predicted scores, not the full $N \times N$ Jacobian of the portfolio. Solve the following adjoint system for $v \in \mathbb{R}^N$ and $q \in \mathbb{R}$:

$$K^\top \begin{bmatrix} v \\ q \end{bmatrix} = \begin{bmatrix} g_w \\ 0 \end{bmatrix}.$$

To see why this works, combine it with the previous differential equation:

$$d\mathcal{L} = \begin{bmatrix} g_w \\ 0 \end{bmatrix}^\top \begin{bmatrix} dw^* \\ d\nu^* \end{bmatrix} = \begin{bmatrix} v \\ q \end{bmatrix}^\top \begin{bmatrix} d\hat{r} \\ 0 \end{bmatrix} = v^\top d\hat{r}.$$

Therefore,

$$\boxed{\frac{\partial \mathcal{L}}{\partial \hat{r}} = v, \quad \nabla_{\theta} \mathcal{L} = \left(\frac{\partial \hat{r}}{\partial \theta} \right)^\top v.}$$

PyTorch passes v through score standardization and the GRU. The optimizer then updates θ . We solve a linear system; we do not explicitly form a matrix inverse.

Backward Pass: How This Is Used in the Implementation

1. Predict $\hat{r}$, then solve the barrier problem for w^* using CVXPY and Clarabel.
2. Evaluate the original decision loss and compute g_w .
3. Build the barrier curvature at w^* and solve for the gradient with respect to $\hat{r}$.
4. Pass this gradient back through the prediction model.

The barrier layer uses an explicit implicit derivative, rather than a `CvxpyLayer` backward call. It differentiates the optimality equations at the solution, without unrolling solver iterations.

The derivation above assumes a strictly feasible point. If a weight is fixed by equal bounds, eliminate it first; its zero slacks cannot enter a logarithm. Keep any remaining equalities as $Aw = b$. For a basis Z of feasible directions ($AZ = 0$), the equivalent solve is

$$(Z^\top HZ)u = Z^\top g_w, \quad \frac{\partial \mathcal{L}}{\partial \hat{r}} = Zu,$$

expressed in the remaining weight coordinates. This derivative is for the current, finite μ and δ ; it is not the derivative of the original nonsmooth problem.

References
